Mohid Tahir

Mechanical Engineer

+1 (647) 778-8676 mohid.tahir@mail.utoronto.ca mohidtahir.ca linkedin.com/in/mohid-tahir

Education

University of Toronto **Sep. 2021 – Apr. 2026**
Bachelor of Applied Science and Engineering *Toronto, ON*

- Program: Mechanical Engineering + PEY Co-op, Minor: Engineering Business, Certificate: EV Design
- Related Coursework: Mechanical Engineering Design, Kinematics and Dynamics of Machines, Electric Vehicle Design, Analog and Digital Electronics, Design for the Environment

Meadowvale Secondary School **Sep. 2017 – Jun. 2021**
Ontario Secondary School Diploma *Mississauga, ON*

Technical Skills and Qualifications

Software & Tools: SolidWorks, AutoCAD, Onshape, Minitab, Microsoft Office Suite (Excel, Word, PowerPoint), Calibre
Programming & Languages: Python, MATLAB, CSS, HTML, Git, PLC, G-code
Simulation & Analysis: Life Cycle Assessment (LCA), Net Present Value (NPV) Modeling, DMAIC, Weighted Decision Matrices, Circuit Simulation (SuperSpice)
Fabrication & CAD Practices: Manual Machining (lathe, mill, drill press), PCB Design, 3D Modeling, Reverse Engineering, Tolerance Analysis
Soft Skills: Time Management, Collaboration, Critical Analysis, Verbal & Written Communication, Technical Documentation, Client Communication, Project Management

Professional Experience

Mechanical Engineering Intern (Special Projects) **May 2024 – Aug. 2025**
AGS Automotive Systems *Oshawa, ON*

- Designed 20+ part and process models in SolidWorks, improving manufacturing efficiency and drawing accuracy
- Updated schematic drawings in AutoCAD to ensure consistency across production projects
- Analyzed time data using the MOST Technique, achieving a 15% reduction in process times for key workflows
- Conducted quality audits on over 50 steel bundles per week, ensuring compliance with company standards
- Performed process improvement analysis in Excel, supporting workflow enhancements and data-driven decision making

Design Engineering Intern **May 2023 – Aug. 2023**
Parts Apart *Toronto, ON*

- Conducted the end-to-end development of a product, overseeing 4 stages of the product development lifecycle
- Performed extensive market research to identify and define a target niche, one that impacts 40% of the population
- Employed computer-aided design (CAD) and manufacturing processes to develop and enhance prototypes
- Collaborated closely with the company founder to transform the concept into a tangible product, costing under $50

Communications Manager & Project Team Member **Jan. 2022 – Apr. 2022**
Engineering Strategies and Practice II (APS112) *Toronto, ON*

- Collaborated with a university professor to design a two-variable teaching device that allowed beginner engineering students to run physical DMAIC experiments
- Utilized functional decomposition, morphological charts, and multivoting to develop an adjustable ramp and continuous ball drop mechanism constrained by classroom safety limits
- Validated the system by running 25 tests across 5 variable combinations, modeling the transfer function in Minitab to prove experimental reliability

Shift Leader **Jun. 2020 – Jan. 2024**
Dairy Queen *Mississauga, ON*

- Led a 6 to 8 person team through weekday rushes, owning cash reconciliation, order accuracy, and escalated customer issues across 3.5 years of part-time work alongside university
- Trained new hires on POS, food safety, and service standards, shortening ramp-up time for each new cohort
- Managed end-of-shift store operations, including register balancing and daily inventory management, ensuring smooth handoffs between shifts
- Actively contributed to new product ideation and promotional rollouts while building team morale in a fast-paced environment

Projects

Low-Speed PMSG for Vertical Axis Wind Turbine **Sep. 2025 – Apr. 2026**
Capstone Design (MIE490/491), Client: UTWind *University of Toronto*

- Designed a custom direct-drive outer-rotor radial-flux permanent magnet synchronous generator for UTWind's 2026 vertical axis wind turbine, targeting 600 to 1200 W output at 100 to 200 RPM with under 0.1 Nm starting torque
- Screened four generator architectures using a weighted decision matrix, eliminating candidates with poor low-RPM efficiency and high cogging torque
- Collaborated with teammates to optimize rotor and stator geometry based on electromagnetic simulation data, successfully hitting 842 W output and 0.09 Nm peak starting torque
- Delivered a fabrication-ready SolidWorks assembly with procurement specs for NdFe35 magnets and ASTM A677 (M19) lamination steel, setting up the 2026 turbine build

The WindRunner **Jan. 2026 – Apr. 2026**
Product Design (MIE540) *University of Toronto*

- Designed a propeller-driven sprint car targeting a $15 retail price and a strict $6 BOM, balancing speed, durability, and a child-safe mesh guard constraint
- Conducted full factorial DOE and factor analysis, identifying an 85 mm wheel diameter as the dominant variable that improved track times by nearly 10 seconds across carpet and tile surfaces
- Performed worst-case and RSS tolerance analysis to finalize component geometry, validating the optimized physical prototype against Voice of Customer requirements

All-Wheel-Drive Demonstration Platform **Sep. 2025 – Dec. 2025**
Electric Vehicle Design (APS380) *University of Toronto*

- Designed and built a scaled dual-motor split-axle AWD platform in SolidWorks, integrating front and rear differentials to demonstrate EV drivetrain architecture and torque distribution
- Instrumented the platform with an ESP32 microcontroller and Hall-effect sensors to capture wheel-speed data and observe traction slip behavior during live test runs
- Iterated on belt and pulley geometry across multiple CAD revisions, resolving critical belt-skip issues and adjusting to 3D printing tolerances after pivoting from an initial planetary gear layout

Printed Circuit Board **Jan. 2024 – Apr. 2024**
Analog & Digital Electronics (MIE346) *University of Toronto*

- Designed a custom printed circuit board to scale a low-voltage analog signal from 0 to 0.35 V to a 0.25 to 3.3 V microcontroller range, incorporating a non-inverting op-amp configuration
- Developed a two-stage Butterworth low-pass filter with a 1 kHz cutoff using the Analog Devices Filter Tool, simulating circuit sweeps in SuperSpice
- Executed schematic capture and PCB layout using Multisim, Ultiboard, and Eagle, successfully fabricating the board and validating less than 4% error against theoretical predictions

Google Home Housing **Jan. 2024 – Apr. 2024**
Design for the Environment (MIE315) *University of Toronto*

- Ran a materials trade study comparing plastic and FSC-certified wood for smart speaker housings, utilizing a weighted decision matrix and Life Cycle Assessment (LCA)
- Modeled a 5-year Net Present Value (NPV) at a 5% rate, demonstrating wood achieved a more favorable NPV (-$82.10 vs -$178.34) alongside lower cumulative emissions
- Delivered a comprehensive consulting recommendation that owned the engineering trade-offs, factoring in machining energy, varnish ecotoxicity, acoustics, and public perception

Robotic Arm for Rover **Sep. 2023 – Dec. 2023**
Kinematics and Dynamics of Machines (MIE301) *University of Toronto*

- Redesigned a sample-collection robotic arm for Io's terrain, adapting the architecture from a rover base to a compact drone-mounted configuration
- Optimized link lengths for a folding four-bar linkage using MATLAB, ensuring the entire assembly packed strictly within a 58.5 cm by 48.9 cm drone envelope
- Modeled the joint stack and frame in SolidWorks, selecting Aluminum Alloy 1060 to maximize the strength-to-weight ratio for aerial deployment

Pick and Place CNC Machine — Jan. 2023 – Apr. 2023
Mechanical Engineering Design (MIE243) — *University of Toronto*
- Conceptualized a desktop CNC pick-and-place machine for educational PCB assembly, focusing on modularity, affordability, and a ± 0.01 inch placement accuracy
- Designed a 3-axis motion system in SolidWorks, integrating a lead screw z-axis with sleeve bearings, interchangeable rotational vacuum heads, and a custom ratchet-indexed reel feed
- Sourced all mechanical components through McMaster-Carr to generate a realistic BOM, and specified a camera-based alignment subsystem to ensure reliable part orientation

Mechanical Dissections — Jan. 2023 – Apr. 2023
Mechanical Engineering Design (MIE243) — *University of Toronto*
- Reverse-engineered physical devices including a jigsaw, can opener, and screwdriver, utilizing calipers to measure parts and rebuild them digitally in SolidWorks
- Modeled a complete 12:1 parallel gearbox from scratch, correctly dimensioning and mating gears, shafts, and housings to simulate functional rotation
- Produced hand-drafted engineering drawings and detailed parametric models, bridging the gap between physical measurement and CAD assembly

Miniature Pneumatic Engine — Nov. 2022
George Brown College Machining — *Toronto, ON*
- Machined a functional pneumatic engine using manual mills, lathes, and drill presses, applying foundational fabrication skills
- Optimized feed rates and cutting speeds to ensure tight dimensional tolerances, safe operation, and high-quality surface finishes

Short Story – "Snow Falling Soundlessly at Night" — Feb. 2020
Meadowvale Secondary School — *Mississauga, ON*
- Authored and formatted an original short story as an EPUB publication for a scholarship submission
- Explored narrative themes of memory and resilience to showcase strong written communication and structured project execution

Handmade Coffee Table & Jewellery Box — Dec. 2018
Meadowvale Secondary School — *Mississauga, ON*
- Designed and assembled functional wooden furniture pieces, utilizing band saws and belt sanders to achieve precise symmetry
- Applied professional finishing techniques and dimensional planning to translate raw materials into project-ready pieces

CO_2 Gas Car & Birdhouse Construction — Oct. 2017 – Dec. 2017
Meadowvale Secondary School — *Mississauga, ON*
- Constructed a functional birdhouse and a CO_2-powered wooden car, applying early measurement and cutting skills
- Utilized foundational engineering principles including aerodynamics, weight distribution, and structural layout to optimize design performance

Volunteering & Extra-Curriculars

Frosh Leader – Skule Frosh Team — Sep. 2022
University of Toronto — *Toronto, ON*
- Led orientation activities for new engineering students, promoting inclusion and engagement across diverse cohorts

Photographer — Nov. 2019 – Mar. 2020
Meadowvale Secondary School — *Mississauga, ON*
- Captured key school and community events for newspaper publication, using photography to document student life

Summer Camp Counselor — Aug. 2019
City of Mississauga — *Mississauga, ON*
- Led orientation activities for youth, promoting inclusion and engagement across diverse cohorts

World of Welcome Volunteer — Aug. 2019
Peel District School Board — *Mississauga, ON*
- Helped immigrant students acclimate to school life in Canada by leading orientation and peer support activities

Event Assistant — Jul. 2019
ICNA Relief Canada — *Mississauga, ON*
- Supported event logistics and merchandise sales for charity events, ensuring smooth setup and breakdown

Awards & Certifications

Dean’s Honours List, Certified SolidWorks Associate (CSWA), Basic Machining (George Brown), Secondary School Valedictorian, Ontario Scholar, Subject Award – International Business (99%), Subject Award – Canadian Law (99%)